

TASK 14: Mini Project – Console-Based Inventory Management System

Tools:

- Primary: IntelliJ IDEA, Java Collections
- Alternatives: Eclipse

Hints / Mini Guide:

1. Design Product class with attributes like id, name, quantity, price.
2. Store products using HashMap.
3. Implement add, update, delete, and view operations.
4. Validate product IDs uniquely.
5. Display inventory summary.
6. Persist data using file handling.
7. Handle invalid user inputs.
8. Refactor code using methods.

Deliverables:

- Inventory management console application
- Inventory data file

Interview Questions Related To Above Task:

- Why HashMap for inventory?
- How to ensure data consistency?
- What happens on duplicate keys?
- How to persist data?
- Scalability issues?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

